SYED ABDUL SAMAD ALI

Computer Systems Engineer | RTL / Digital IC Design

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PROFESSIONAL SUMMARY

Computer Systems Engineering graduate with hands-on experience in Digital Logic Design, Verilog RTL development, hardware protocols, computer architecture, and embedded systems. Currently participating in an intensive PSEB/MERL upskilling program focused on Digital IC Design, RTL development, and semiconductor engineering workflows. Building foundational knowledge in SystemVerilog and ASIC Design Flow. Experienced with QuestaSim, GTKWave, Logisim, Intel Quartus, and Ubuntu for RTL simulation, debugging, waveform analysis, and hardware design workflows. Strong foundation in C/C++, FSM-based design, AMBA APB, UART, SRAM, cache architectures, and memory systems.

TECHNICAL SKILLS

- **HDL & Programming:** Verilog HDL, C/C++, Python
- **Digital & RTL Design:** Digital Logic Design, RTL Design, FSMs, Computer Architecture, Cache Controllers, SRAM, Memory Controllers
- **Protocols & Interfaces:** AMBA APB, UART, LoRa
- **Simulation & Design Tools:** QuestaSim, GTKWave, Logisim, Intel Quartus, VS Code
- **Embedded Platforms:** ESP32, Raspberry Pi
- **Operating Systems:** Ubuntu, Windows
- **Currently Learning:** SystemVerilog, ASIC Design Flow

PROFESSIONAL EXPERIENCE

Digital IC Design Trainee

Micro Electronics Research Lab (MERL) / PSEB — Karachi, Pakistan | May 2026 – Present

- Participating in a PSEB industry upskilling program focused on Digital IC Design, RTL development, and semiconductor engineering workflows.
- Developed and simulated digital logic and Verilog RTL designs, strengthening practical understanding of combinational/sequential logic and FSM-based architectures.
- Worked with hardware interfaces and protocols including AMBA APB and UART, developing RTL models for protocol communication and control.
- Used QuestaSim and GTKWave in Ubuntu environments for simulation, debugging, waveform inspection, and verification of digital designs.
- Applied C/C++ to hardware-oriented programming, register-level control, and logic modeling exercises.
- Currently building foundational knowledge of SystemVerilog and ASIC Design Flow as part of continued semiconductor-focused training.

Design Engineering Intern

Xclusive Solutions (PVT) LTD. — Karachi, Pakistan | Dec 2025 – Jan 2026

- Designed Fire Alarm and Life Safety System layouts for a 50,000+ sq. ft. industrial facility using AutoCAD.
- Assisted engineering teams in preparing technical drawings according to project specifications and NFPA 72 requirements.
- Coordinated drawing details and engineering documentation to support project implementation.

ENGINEERING PROJECTS

Cache Controller & VGA Display via SRAM

Verilog | QuestaSim | GTKWave | Intel Quartus | SRAM | RTL Design

- Designed and implemented a Verilog RTL cache controller integrated with a single-port SRAM architecture.
- Developed a VGA display interface and integrated memory access logic within the RTL design.
- Implemented and analyzed memory read/write operations, control logic, and timing behavior through simulation.
- Used QuestaSim and GTKWave to debug RTL behavior and inspect simulation waveforms.
- Prepared FPGA-oriented design constraints using Intel Quartus.

AMBA APB Interface & UART Controller

Verilog | Logisim | C++ | AMBA APB | UART

- Developed RTL models for AMBA APB Master/Slave interfaces, implementing FSM-based handshaking and register access logic.
- Designed and simulated a UART controller supporting TX, RX, parity, and control/status functionality.
- Built a gate-level UART model in Logisim to strengthen understanding of digital logic implementation.
- Developed C++ firmware modules for UART register control and status monitoring.

- Analyzed protocol behavior and digital timing through simulation and functional testing.

Interactive Parameterized Cache Simulator

C++ | Dear ImGui | Computer Architecture

- Developed a parameterized cache simulator to analyze cache replacement policies and memory hierarchy behavior.
- Implemented configurable cache parameters and replacement strategies including FIFO, LRU, and Random.
- Built an interactive visualization interface using Dear ImGui for analyzing cache behavior and simulation results.
- Strengthened understanding of memory management, data structures, cache organization, and computer architecture.

Real-Time Secure Communication System — Final Year Project

Raspberry Pi | LoRa | Python | AES Encryption

- Led a three-member team in developing a secure long-range wireless communication system using Raspberry Pi and LoRa transceivers.
- Implemented AES encryption in Python to protect transmitted data.
- Integrated embedded hardware and software components for real-time wireless communication.
- Developed and tested the system for reliable long-range data transmission.

EDUCATION

Bachelor of Engineering (Computer Systems)

UIT University — Karachi, Pakistan | Graduated: July 2026

Relevant Coursework: Computer Architecture, Digital Logic Design, Embedded Systems, Computer Networks, Internet of Things

CERTIFICATIONS & TRAINING

- PSEB / MERL — Digital IC Design Industry Upskilling Program (May 2026 – Present)
- NED University — IC Design Summer School (2024)
- IBM (Coursera) — Introduction to Data Analytics